

**Question 1: Object-Oriented Programming (OOP)****[8 marks]****Introduction: (2 marks)**

Object-Oriented Programming (OOP) is a programming paradigm that structures software design around objects rather than functions and logic. An object is a self-contained entity that encapsulates both

**Core Concepts: (3 marks)**

OOP is founded on four core pillars. Encapsulation bundles data and methods within a class and restricts direct external access to internal state using access modifiers (private, protected, public), exposing only a controlled interface through getter/setter methods · this prevents unintended data corruption. Abstraction hides

**Examples: (2 marks)**

A BankAccount class encapsulates balance (private) and exposes deposit(amount) and withdraw(amount) methods · Encapsulation. SavingsAccount extends BankAccount and overrides calculateInterest() to add 4% per annum · Inheritance and Polymorphism. An

**Conclusion: (1 mark)**

OOP provides a robust, scalable approach to software development by modelling real-world entities as objects. Its

**Question 2: Database Normalization****[8 marks]****Introduction: (2 marks)**

Database normalization is a systematic process of organizing a relational database schema to minimize data redundancy and eliminate update, insert, and delete anomalies. It involves decomposing larger tables into

**Core Concepts: (3 marks)**

First Normal Form (1NF): each column must contain atomic (indivisible) values, every row must be unique (primary key exists), and there must be no repeating groups or arrays within a column. Second Normal Form (2NF): the table must

**Examples: (2 marks)**

Consider Student\_Course(StudentID, StudentName, CourseID, CourseName, InstructorName, Grade). Partial dependency: StudentName depends only on StudentID (violates 2NF). Transitive dependency: InstructorName depends on CourseID,

**Conclusion: (1 mark)**

Normalization produces clean, anomaly-free database schemas by systematically removing redundancy. While 3NF or

**Question 3: Operating System Process Management****[8 marks]****Introduction: (2 marks)**

Process management is a core responsibility of the operating system (OS), which must create, schedule, synchronize, and terminate processes while ensuring CPU utilization, fairness, and system stability. A process is an

**Core Concepts: (3 marks)**

Process States: New · Ready · Running · Waiting/Blocked · Terminated. A process moves between states based on scheduling decisions and I/O events. PCB stores: PID, state, program

counter, CPU registers, memory limits, open

**Examples: (2 marks)**

Round Robin with quantum=4ms on P1(24ms), P2(3ms), P3(3ms): execution order P1(4), P2(3), P3(3), P1(4)×5. Average waiting time =  $(6+4+7)/3 = 5.67$ ms. Context switch example:

**Conclusion: (1 mark)**

Effective process management is the cornerstone of OS performance. The scheduler must balance CPU

**Question 4: Computer Networks and the OSI Model**

[8 marks]

**Introduction: (2 marks)**

A computer network is a collection of interconnected devices (hosts, routers, switches) that communicate by exchanging data packets according to agreed-upon protocols. Networks range from LANs

**Core Concepts: (3 marks)**

Layer 7 · Application: provides network services directly to end-user applications. Protocols: HTTP/HTTPS (web), SMTP/IMAP/POP3 (email), FTP/SFTP (file transfer), DNS (name resolution), DHCP (address allocation), SNMP (network management). Layer 6 · Presentation: data translation (encoding/decoding), encryption/decryption

**Examples: (2 marks)**

HTTPS request to www.example.com: DNS resolves domain · IP (Application). TLS encrypts data (Presentation). TCP session via 3-way handshake (Session + Transport).

**Conclusion: (1 mark)**

The OSI model is the definitive reference for understanding, designing, and troubleshooting network

**Question 5: Data Structures · Binary Trees and BST**

[8 marks]

**Introduction: (2 marks)**

A tree is a non-linear hierarchical data structure composed of nodes connected by edges, with a designated root node and no cycles. A binary tree is a

**Core Concepts: (3 marks)**

Binary Tree types: Full (every node has 0 or 2 children), Complete (all levels fully filled except possibly the last, which is filled left-to-right), Perfect (all internal nodes have 2 children, all leaves at same level), Degenerate/Skewed (each

**Examples: (2 marks)**

BST insertion of [50, 30, 70, 20, 40, 60, 80]: root=50, left=30, right=70, 30's left=20, 30's right=40, 70's left=60, 70's

**Conclusion: (1 mark)**

Binary trees and BSTs provide efficient  $O(\log n)$  search, insert, and delete

**Question 6: Sorting Algorithms and Algorithm Analysis**

[8 marks]

**Introduction: (2 marks)**

Sorting is the process of arranging elements in a defined order (ascending or descending). It is

one of the most fundamental operations in computer science, underpinning database query

**Core Concepts: (3 marks)**

Bubble Sort: repeatedly swaps adjacent out-of-order elements. Best  $O(n)$  (already sorted, with flag), Worst/Average  $O(n^2)$ . Stable, in-place. Selection Sort: finds minimum element and places at front each pass. Always  $O(n^2)$ . Unstable, in-place. Insertion Sort:

**Examples: (2 marks)**

Quick Sort on [3,6,8,10,1,2,1] with pivot=1: partition gives [1,1,3,6,8,10,2] · recurse on left/right halves. Merge

**Conclusion: (1 mark)**

Choosing the right sorting algorithm depends on input size, data characteristics (nearly

**Question 7: Software Engineering · SDLC and Agile**

**[8 marks]**

**Introduction: (2 marks)**

Software Engineering applies systematic, disciplined, quantifiable approaches to the development, operation, and maintenance of software · adapting engineering principles to software systems. The Software Development Life

**Core Concepts: (3 marks)**

Classical SDLC Models: Waterfall · sequential phases (Requirements · Design · Implementation · Testing · Deployment · Maintenance); simple but inflexible for changing requirements; suits well-understood, stable domains. V-Model · extends Waterfall with verification and validation

**Examples: (2 marks)**

Waterfall for avionics software: requirements fully frozen, strict V&V, DO-178C certification requires documented phase gates · Waterfall is appropriate. Scrum for e-commerce platform: Sprint 1: user

**Conclusion: (1 mark)**

Modern software engineering favours Agile methodologies for their ability to accommodate changing requirements and deliver

**Question 8: Computer Architecture · CPU and Memory Hierarchy**

**[8 marks]**

**Introduction: (2 marks)**

Computer architecture defines the functional structure and behaviour of a computer system at the ISA (Instruction Set Architecture) level · the interface

**Core Concepts: (3 marks)**

CPU Components: ALU (Arithmetic Logic Unit · performs arithmetic and logical operations), CU (Control Unit · fetches, decodes, and issues micro-operations), Registers (PC/IP, IR, MAR, MDR, accumulator, general-purpose). Instruction Cycle: Fetch (load instruction from

**Examples: (2 marks)**

5-stage pipeline example: ADD R1,R2,R3 followed by SUB R4,R1,R5 causes RAW hazard on R1 · solved by

**Conclusion: (1 mark)**

Modern CPUs achieve high performance through deep pipelines (10-20 stages), out-of-order execution, superscalar issue

**Question 9: Web Technologies · Client-Server Architecture and HTTP****[8 marks]****Introduction: (2 marks)**

Web technologies encompass the protocols, languages, and frameworks used to build and deliver content over the World Wide Web. The client-server architecture is the foundational model of the web:

**Core Concepts: (3 marks)**

HTTP Request/Response: method (GET, POST, PUT, DELETE, PATCH, HEAD, OPTIONS), URL, headers (Content-Type, Authorization, Cache-Control, Accept), body (for POST/PUT). Status codes: 2xx Success (200 OK, 201 Created, 204 No Content), 3xx Redirection (301 Moved Permanently, 302 Found), 4xx

**Examples: (2 marks)**

GET /api/users/123 HTTP/1.1 · 200 OK with JSON body {id:123, name:'Alice'}. POST /api/login with {email, password} · server validates · returns

**Conclusion: (1 mark)**

Modern web architecture has evolved from simple static pages to complex

**Question 10: Cloud Computing · Service Models and Deployment****[8 marks]****Introduction: (2 marks)**

Cloud computing delivers computing resources (servers, storage, databases, networking, software, analytics) over the internet ('the cloud') on an on-demand, pay-as-you-go model, eliminating the need for organizations to own

**Core Concepts: (3 marks)**

Service Models · IaaS (Infrastructure as a Service): provides virtualized compute, storage, and networking. Customer manages OS, middleware, apps. Providers: AWS EC2/S3, Azure VMs, Google Compute Engine. Use case: hosting custom applications, dev/test environments. PaaS (Platform as a Service): provides a development platform with

**Examples: (2 marks)**

Netflix uses AWS IaaS (EC2, S3, CloudFront CDN) with multi-region deployment for 99.99% availability. A startup uses Heroku PaaS

**Conclusion: (1 mark)**

Cloud computing has democratized access to enterprise-grade infrastructure, enabling startups to compete with large enterprises. Choosing

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